MECH559: Term Project Proposal

Optimization of Log-Yard Operations Using Discrete-Event Simulation

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Abstract

A log yard is the part of a wood-processing plant where trucks are unloaded, logs are stored by species, and loaders move logs to the mill infeed. The same loaders are needed to unload waiting trucks and to keep the mill supplied. This creates a scheduling problem: serving one request can delay another. The proposed project will build a discrete-event simulation of this system in Python using SimPy. The first study will use fixed truck-arrival and mill-demand schedules. An interpretable dispatch policy will assign available loaders to feasible material-transfer tasks using four weights for truck urgency, mill urgency, inventory balance, and operating cost. These weights will be optimized for several schedule-demand pairs. The next study will add loader count and pile locations as physical design variables. The final planned study will make truck arrivals and truck contents stochastic and will evaluate both the mean and variance of shift performance. The main performance measures are truck waiting time, mill production shortfall, and yard cost, including inventory holding cost. The baseline task model allows transfers from trucks to piles, piles to the infeed, and trucks directly to the infeed. Optional extensions will examine multistage routes, task interruption, short-horizon future-cost estimates, equipment failures, and age-dependent inventory quality.

Contents

1 Introduction **4**

2 Component Design **4**

2.1 Inbound Material Management – Jessica Mizrahi 5

2.1.1 Problem Statement 5

2.1.2 Nomenclature 5

2.1.3 Mathematical Formulation 6

2.2 Loader Fleet – Lucas Bessai 7

2.2.1 Problem Statement 8

2.2.2 Nomenclature 8

2.2.3 Mathematical Formulation 9

2.3 Mill Infeed and Downstream Demand – Benjamin Jackler 10

2.3.1 Problem Statement 10

2.3.2 Nomenclature 10

2.3.3 Mathematical Formulation 11

2.4 Shared Variables and Component Coupling 12

3 Staged Optimization Framework **13**

3.1 Simulation Evaluation 13

3.2 Stage 1: Deterministic Schedules and Policy Weights 13

3.3 Stage 2: Physical Design Variables 14

3.4 Stage 3: Stochastic Truck Supply 15

3.5 Integrated Multi-objective Problem in Negative-Null Form 15

4 Dispatch Policy **17**

4.1 Tasks and Assignment Rule 18

4.2 Four Policy Variables 19

4.2.1 Feature Parameters and Derived Quantities 19

4.3 Initial Feature Definitions 20

4.3.1 Truck Urgency 20

4.3.2 Mill Urgency 20
4.3.3 Inventory Balance 21
4.3.4 Operating Cost 22
4.4 Example Dispatch Decision 22
4.5 Idle Decision 23

5 Conditional Extensions **23**

5.1 Extension Nomenclature 23
5.2 Production Backlog and Backlog-Aware Dispatch 24
5.3 Multistage Routes and Task Preemption 26
5.4 Expected Future Cost Rollout 27
5.5 Breakdowns and Robustness Sensitivity 27
5.6 Ageing Inventory and Quality Loss 28

1 Introduction

A log yard connects inbound transportation to mill production. Trucks arrive with one or more log species. Loaders unload the trucks, build species-specific piles, and replenish a finite mill infeed. A loader assigned to an arriving truck reduces the truck queue but may leave the mill without the required species. A loader assigned to the mill protects production but may increase truck waiting. Larger inventories provide a buffer against uncertain supply, but inventory occupies yard space, requires handling, and has a holding cost.

The project will model these interactions using discrete-event simulation. The simulated clock advances from one event to the next, such as a truck arrival, loader-task completion, mill-demand change, or inventory depletion. SimPy provides the environment, processes, events, and shared-resource mechanisms needed for this type of model [2].

The study is divided into three components: inbound material management, the loader fleet, and the mill infeed and downstream demand. Each component has a local performance measure, but the components share loader assignments and material inventories. Improving each component independently may therefore fail to improve the complete yard. The final design must be evaluated through the integrated simulation.

2 Component Design

This section defines the three component models, their responsibilities, objectives, variables, parameters, and constraints. The component assignments are shown in Table 1.

Table 1: Component responsibilities.

Team member	Component	Main responsibility
Jessica Mizrahi	Inbound material management	Truck arrivals, truck queues, unloading positions, truck contents, and pile entry.
Lucas Bessai	Loader fleet	Loader availability, travel, task duration, assignment feasibility, dispatch policy, and operating cost.
Benjamin Jackler	Mill infeed and downstream demand	Species-specific demand, infeed inventory, actual production, and production shortfall.

2.1 Inbound Material Management – Jessica Mizrahi

This component describes how trucks enter the yard, wait for service, and transfer material to piles or directly to the mill infeed.

2.1.1 Problem Statement

Truck b arrives at time a_b with volume v_{bs} of species s. An available loader may transfer a fixed batch from the truck to the matching species pile or, when yard rules and infeed capacity permit, directly to the infeed. Direct transfer can reduce travel, handling, and short-term storage. Pile storage can preserve supply for later demand and allows older pile inventory to be consumed before new material. The baseline model includes both routes, while a facility parameter can disable direct truck-to-infeed service if it is not physically allowed.

The local objective is to reduce truck waiting. The component must also enforce truck-content, unloading-position, and pile-capacity limits.

2.1.2 Nomenclature

Symbol	Definition	Units
$\mathcal{S}$	Set of log species, indexed by s	–
$\mathcal{B}$	Set of trucks, indexed by b	–
$\mathcal{M}$	Set of candidate pile sites, indexed by m	–
a_b	Arrival time of truck b	min
$v_{bs}(t)$	Species-s volume remaining on truck b	m^3
n_B	Number of unloading positions	integer
$I_s^P(t)$	Inventory in the pile for species s	m^3
C_s^P	Fixed pile capacity	m^3
$I_s^F(t)$	Infeed inventory of species s	m^3
C_s^F	Fixed infeed capacity for species s	m^3
N_T	Number of trucks in the simulated shift	integer
P_s	Pile node for species s	yard node
F_s	Mill-infeed node for species s	yard node
$R_s^{P,\mathrm{comm}}(t)$	Volume of species s committed to enter its pile	m^3
$R_s^{F,\mathrm{comm}}(t)$	Volume of species s committed to enter the mill infeed	m^3
q	Fixed material-transfer batch size	m^3

Symbol	Definition	Units
t_k	Time of simulation event k	min
W_b	Waiting time before truck b begins service	min
$W_{0.95}$	95th percentile of truck waiting time	min
$\overline{W}_{0.95}$	Maximum allowable 95th-percentile truck waiting time	min
y_{sm}	1 if the pile for species s is assigned to site m, 0 otherwise	0 or 1
m_s	Candidate site assigned to the pile for species s	integer
χ_{TF}	Direct truck-to-infeed permission parameter	0 or 1

2.1.3 Mathematical Formulation

A truck-to-pile task $j = (b, P_s, s, q)$ is feasible if

$$q - v_{bs}(t_k) \leq 0, \tag{1}$$
$$I_s^P(t_k) + R_s^{P,\text{comm}}(t_k) + q - C_s^P \leq 0. \tag{2}$$

A direct truck-to-infeed task $j = (b, F_s, s, q)$ additionally requires

$$q - v_{bs}(t_k) \leq 0, \tag{3}$$
$$I_s^F(t_k) + R_s^{F,\text{comm}}(t_k) + q - C_s^F \leq 0, \tag{4}$$
$$1 - \chi_{TF} \leq 0. \tag{5}$$

The transfer updates the truck inventory and the inventory at the destination node $d(j)$, where $I_{d(j),s}$ denotes the species-s inventory at that destination (either the corresponding pile or the mill infeed):

$$v_{bs}(t_k^+) = v_{bs}(t_k^-) - q, \tag{6}$$
$$I_{d(j),s}(t_k^+) = I_{d(j),s}(t_k^-) + q. \tag{7}$$

When several loaders are assigned at the same event, source inventories and truck volumes are evaluated after subtracting material reserved by earlier assignments. Destination-capacity checks include the corresponding committed incoming volumes.

The local objective is the mean truck wait,

$$W_{\text{truck}} = \frac{1}{N_T} \sum_{b=1}^{N_T} \left(t_b^{\text{start}} - a_b \right). \tag{8}$$

In the second optimization stage, each species pile is assigned to one candidate yard site. If $y_{sm} = 1$ means that species s uses site m, then

$$\sum_{m \in \mathcal{M}} y_{sm} - 1 = 0 \qquad \forall s, \tag{9}$$

$$\sum_{s \in \mathcal{S}} y_{sm} - 1 \leq 0 \qquad \forall m, \tag{10}$$

$$y_{sm} \in \{0, 1\}. \tag{11}$$

The selected pile location is $m_s = \sum_m m y_{sm}$. Site choices change loader travel distances but not pile capacities, which remain fixed parameters in the initial study.

The inbound component is summarized in negative-null form as

$$\begin{aligned} \underset{\theta,\, \{y_{sm}\}}{\text{minimize}} \quad & f_T = \frac{W_{\text{truck}}}{W_{\text{ref}}} \\ \text{subject to} \quad & -I_s^P(t_k) \leq 0, \\ & I_s^P(t_k) - C_s^P \leq 0, \\ & q - v_{bs}(t_k) \leq 0, \\ & W_{0.95} - \overline{W}_{0.95} \leq 0, \\ & \sum_m y_{sm} - 1 = 0, \\ & \sum_s y_{sm} - 1 \leq 0. \end{aligned} \tag{12}$$

The dependence on θ is implicit: θ changes the sequence of truck and mill tasks, which changes every truck service time in Eq. (8).

2.2 Loader Fleet – Lucas Bessai

This component represents the shared loaders and the dispatch decisions that connect every source to every permitted destination.

2.2.1 Problem Statement

The initial model uses single-species, fixed-batch, source-to-destination tasks. A task is complete only after travel to the source, loading, travel to the destination, and unloading. Tasks are not interrupted in the baseline. An available loader may serve a truck-to-pile, pile-to-infeed, or truck-to-infeed request. The loader component trades off travel and equipment cost against the ability to respond quickly to both trucks and the mill.

The first study keeps the fleet configuration fixed and optimizes only the dispatch weights. The second study adds the number of loaders as an integer design variable. Increasing the fleet can reduce waiting and shortfall, but it raises the amortized cost per shift.

2.2.2 Nomenclature

Symbol	Definition	Units
$\mathcal{L}$	Set of loaders, indexed by ℓ	–
N_L	Number of loaders	integer
$A_{\ell,k}$	1 if loader ℓ is available at event k	0 or 1
$\rho_{\ell,k}$	Loader location at event k	yard node
Q_ℓ	Loader carrying capacity	m^3
$z_{\ell j,k}$	1 if loader ℓ begins task j at event k	0 or 1
$\tau_{\ell j,k}$	Duration of task j for loader ℓ	min
$\delta_{\ell j,k}$	Loader travel distance for task j	m
θ	Four dispatch-policy weights	–
v_ℓ	Travel speed of loader ℓ	m/min
r_ℓ^{load}	Loading rate of loader ℓ	m^3/min
r_ℓ^{unload}	Unloading rate of loader ℓ	m^3/min
$o(j)$	Source node of task j	–
$d(j)$	Destination node of task j	–
$d(i_1, i_2)$	Travel distance between yard nodes i_1 and i_2	m
L_ℓ^{dist}	Total distance travelled by loader ℓ during the shift	m
T_ℓ^{active}	Total active operating time of loader ℓ	min
c_L	Cost per loader per shift	\$/shift
c_D	Loader travel cost coefficient	\$/m
c_A	Loader active-time cost coefficient	\$/min
h_s	Holding-cost coefficient for species s	$\$/(m^3 \cdot \text{min})$

Symbol	Definition	Units
H	Simulated shift duration	min

2.2.3 Mathematical Formulation

One loader can perform at most one task, and one task can be assigned to at most one loader:

$$\sum_{j \in \mathcal{J}(x_k)} z_{\ell j,k} - A_{\ell,k} \leq 0 \qquad \forall \ell, k, \tag{13}$$

$$\sum_{\ell \in \mathcal{L}} z_{\ell j,k} - 1 \leq 0 \qquad \forall j, k, \tag{14}$$

$$q - Q_\ell \leq 0 \qquad \forall \ell. \tag{15}$$

For a direct task from source $o(j)$ to destination $d(j)$,

$$\tau_{\ell j,k} = \frac{d(\rho_{\ell,k}, o(j))}{v_\ell} + \frac{q}{r_\ell^{\text{load}}} + \frac{d(o(j), d(j))}{v_\ell} + \frac{q}{r_\ell^{\text{unload}}}. \tag{16}$$

Pile-site variables affect both distances in Eq. (16). Once a task begins, $A_{\ell,k} = 0$ until the task finishes.

The baseline yard cost is

$$C_{\text{yard}} = c_L N_L + c_D \sum_{\ell \in \mathcal{L}} L_\ell^{\text{dist}} + c_A \sum_{\ell \in \mathcal{L}} T_\ell^{\text{active}} + \sum_{s \in \mathcal{S}} h_s \int_0^H I_s^P(t)\, dt. \tag{17}$$

The last term is the pile-inventory holding cost. The coefficient h_s may represent occupied space, tied-up material, inspection, and rehandling. Direct truck-to-infeed tasks avoid this term for the volume that bypasses the pile, but they may leave previously stored material in the yard.

The loader component is summarized in negative-null form as

$$
\begin{aligned}
\underset{N_L,\theta,\{y_{sm}\}}{\text{minimize}} \quad & f_L = \frac{C_{\text{yard}}}{C_{\text{ref}}} \\
\text{subject to} \quad & N_L^{\min} - N_L \le 0, \\
& N_L - N_L^{\max} \le 0, \\
& N_L \in \mathbb{Z}, \\
& \sum_j z_{\ell j,k} - A_{\ell,k} \le 0, \\
& \sum_\ell z_{\ell j,k} - 1 \le 0, \\
& q - Q_\ell \le 0.
\end{aligned}
\tag{18}
$$

Again, C_{yard} depends on θ because the policy determines travel, active time, direct bypasses, and the pile-inventory trajectory.

2.3 Mill Infeed and Downstream Demand – Benjamin Jackler

This component converts the demand program and infeed inventory into actual mill production and production shortfall. The baseline mill either operates at its specified processing rate or is stopped by a lack of material.

2.3.1 Problem Statement

The demand program specifies when species s is required. During an active period, the mill processes at a fixed species-specific rate $\bar{r}_s$; outside that period, the requested rate is zero. Thus, $D_s(t)$ takes values in $\{0, \bar{r}_s\}$. If the required species is unavailable at the infeed, the mill is stopped rather than operated at a reduced rate. Both pile-to-infeed and truck-to-infeed tasks can replenish the infeed.

The baseline demand program activates at most one species at a time.

The local objective is to minimize total production shortfall while respecting finite infeed capacity. Infeed capacities are fixed parameters in the planned study rather than optimized design variables.

2.3.2 Nomenclature

Symbol	Definition	Units
$I_s^F(t)$	Infeed inventory of species s	m^3
C_s^F	Fixed infeed capacity for species s	m^3

Symbol	Definition	Units
$D_s(t)$	Species-specific requested processing rate, equal to 0 or $\bar{r}_s$	m^3/min
$\bar{r}_s$	Fixed processing rate when species s is active	m^3/min
$r_s^{\text{proc}}(t)$	Actual processing rate, equal to 0 or $\bar{r}_s$	m^3/min
$r_s^{P\to F}(t)$	Material transfer rate from pile to infeed for species s	m^3/min
$r_s^{T\to F}(t)$	Material transfer rate from truck to infeed for species s	m^3/min
U_{mill}	Total production shortfall	m^3
$\overline{U}_{\text{mill}}$	Maximum allowable unmet demand	m^3
H	Simulated shift duration	min

2.3.3 Mathematical Formulation

The infeed material balance is

$$\frac{dI_s^F(t)}{dt} = r_s^{P\to F}(t) + r_s^{T\to F}(t) - r_s^{\text{proc}}(t), \qquad 0 \le I_s^F(t) \le C_s^F. \tag{19}$$

The baseline mill operates at a static rate when it is supplied:

$$r_s^{\text{proc}}(t) \in \{0, \bar{r}_s\}, \qquad r_s^{\text{proc}}(t) = \bar{r}_s \iff \left(D_s(t) = \bar{r}_s \text{ and } I_s^F(t) > 0\right). \tag{20}$$

The single-mill restriction is

$$\sum_{s\in\mathcal{S}} \mathbb{I}_{\{r_s^{\text{proc}}(t)>0\}} \le 1. \tag{21}$$

The total production shortfall is

$$U_{\text{mill}} = \sum_{s\in\mathcal{S}} \int_0^H \left[D_s(t) - r_s^{\text{proc}}(t)\right] dt. \tag{22}$$

The mill component is summarized in negative-null form as

$$
\begin{aligned}
\underset{\theta}{\text{minimize}} \quad & f_M = \frac{U_{\text{mill}}}{U_{\text{ref}}} \\
\text{subject to} \quad & -I_s^F(t) \le 0, \\
& I_s^F(t) - C_s^F \le 0, \\
& -r_s^{\text{proc}}(t) \le 0, \\
& r_s^{\text{proc}}(t) \in \{0, \bar{r}_s\}, \\
& \sum_{s \in \mathcal{S}} \mathbb{I}_{\{r_s^{\text{proc}}(t) > 0\}} \le 1, \\
& U_{\text{mill}} - \overline{U}_{\text{mill}} \le 0.
\end{aligned}
\tag{23}
$$

The objective depends on θ because the policy determines whether and when the required species reaches the infeed.

2.4 Shared Variables and Component Coupling

This subsection explains why the three component models cannot be evaluated independently even though each has a separate local objective.

At event k, the integrated state contains truck contents and waiting times, pile inventories and locations, loader locations and availability, infeed inventories, and the current demand:

$$
x_k = \left[x_k^{\text{Truck}},\ x_k^{\text{Pile}},\ x_k^{\text{Loaders}},\ x_k^{\text{Infeed}},\ D(t_k) \right]. \tag{24}
$$

The policy maps this shared state to a vector of loader assignments,

$$
u_k = \pi(x_k; \theta), \tag{25}
$$

and the simulation advances according to

$$
x_{k+1} = F(x_k, u_k; d, \zeta), \tag{26}
$$

where d contains physical design variables and ζ identifies the truck and demand scenario. Repeated decisions generate the trajectory

$$
\boldsymbol{x}_0 \to \boldsymbol{x}_1 \to \cdots \to \boldsymbol{x}_K. \tag{27}
$$

The component costs are functions of this entire trajectory:

$$
(W_{\text{truck}}, U_{\text{mill}}, C_{\text{yard}}) = G\left(x_0, x_1, \ldots, x_K\right) = G(d, \theta; \zeta). \tag{28}
$$

This is how θ enters all three component objectives even though it does not appear explicitly in Eqs. (8), (17), or (22). Changing θ changes u_k; changing u_k changes the event sequence and state trajectory; the new trajectory changes every objective.

A feedback-based subsystem solution is possible as a heuristic. For example, the mill component could communicate a species-shortage penalty, the inbound component could communicate a truck-delay penalty, and the loader component could update its dispatch weights. The components could repeat this process until the shared signals stop changing. This may help initialize the search or explain tradeoffs. It does not guarantee the integrated optimum because all components compete for the same discrete loader assignments, and a small assignment change can alter the order of later events. The final comparison will therefore use the integrated simulation, while subsystem feedback may be used for interpretation or search initialization.

3 Staged Optimization Framework

This section describes the order in which the simulation and optimization will be developed. The stages add uncertainty and physical design variables only after the dispatch model has been explored in a deterministic setting.

3.1 Simulation Evaluation

The SimPy environment contains processes for truck arrivals, loader tasks, and mill consumption. Shared state objects store truck contents, pile inventory, infeed inventory, and loader status. Each transfer task uses the same source–destination data structure, so the execution process can support truck-to-pile, pile-to-infeed, and direct truck-to-infeed tasks without separate dispatch logic. A dispatcher is called whenever an event creates an available loader or changes the feasible task set. The simulation returns truck wait, mill shortfall, travel, active time, and inventory histories.

Verification will begin with limiting cases: no arrivals, unlimited infeed inventory, one-loader first-come-first-served operation, and conservation of total material. These cases make errors visible before optimization begins.

3.2 Stage 1: Deterministic Schedules and Policy Weights

Let

$$\zeta = (\mathcal{A}, D) \tag{29}$$

denote one fixed pair consisting of a complete truck schedule $\mathcal{A}$ and demand program D. Arrival times, truck contents, task times, and demand are fixed. For this first stage, physical design d is

also fixed. The only optimized variables are

$$\theta = [\theta_T, \theta_M, \theta_I, \theta_C]^T, \qquad \theta_r \geq 0, \qquad \sum_r \theta_r = 1. \tag{30}$$

There are four weights but three independent degrees of freedom because of the normalization constraint. For a fixed Pareto preference λ, the deterministic policy problem is

$$\theta^*_{\zeta,\lambda} = \arg\min_{\theta \in \Delta_4} J_\lambda(\theta; \zeta). \tag{31}$$

Here, Δ_4 denotes the four-dimensional nonnegative simplex defined by the weight constraints in Eq. (30).

The same calculation will be repeated for a set of truck-schedule and demand-program pairs. Comparing the resulting θ^* values will show whether one policy performs well across scenarios or whether different operating conditions require different preferences.

With a fully deterministic scenario, the complete future event tree is fixed. In principle, one could optimize every action directly:

$$\boldsymbol{u}^*_{0:K-1} = \arg\min_{\boldsymbol{u}_{0:K-1} \in \mathcal{U}(\zeta)} J_\lambda\left(\boldsymbol{u}_{0:K-1}; \zeta\right). \tag{32}$$

This gives a best sequence for that known day and does not require θ or state features, ϕ. It is useful as an offline benchmark, but it is not a deployable policy because it uses full knowledge of future arrivals. The parameterized policy is more useful because it is interpretable and maps any observed state to an action.

3.3 Stage 2: Physical Design Variables

After the policy model is working, the physical design vector becomes

$$d = [N_L, \{y_{sm}\}_{s \in \mathcal{S}, m \in \mathcal{M}}]. \tag{33}$$

The number of loaders and pile-site assignments are optimized jointly with θ:

$$(d^*, \theta^*) = \arg\min_{d,\theta} J_\lambda(d, \theta; \zeta). \tag{34}$$

These variables are expected to produce clearer system-level tradeoffs than capacity optimization. Loader count directly trades capital cost against delay, while pile locations change travel time for truck unloading and mill replenishment. Pile and infeed capacities remain fixed parameters and

can be varied in sensitivity studies.

3.4 Stage 3: Stochastic Truck Supply

In the final planned stage, truck arrival times and truck contents are random. Let ξ denote one sampled realization. For fixed (d, θ), the scalarized shift cost

$$J_\lambda(d, \theta; \xi) \tag{35}$$

is now a random variable rather than a single value. Its mean and variance are

$$\mu_J(d, \theta) = \mathbb{E}_\xi[J_\lambda(d, \theta; \xi)], \tag{36}$$

$$\sigma_J^2(d, \theta) = \mathrm{Var}_\xi[J_\lambda(d, \theta; \xi)]. \tag{37}$$

The stochastic problem is

$$\underset{d,\theta}{\text{minimize}} \quad \begin{bmatrix} \mu_J(d, \theta) \\ \sigma_J^2(d, \theta) \end{bmatrix}. \tag{38}$$

The mean measures average performance. The variance measures consistency across possible days. In plots and tables, standard deviation may be reported because it has the same units as J_λ.

The expectations will be estimated using R replications:

$$\widehat{\mu}_J = \frac{1}{R} \sum_{r=1}^{R} J_\lambda(d, \theta; \xi_r), \tag{39}$$

$$\widehat{\sigma}_J^2 = \frac{1}{R-1} \sum_{r=1}^{R} \left[J_\lambda(d, \theta; \xi_r) - \widehat{\mu}_J \right]^2. \tag{40}$$

Common random scenarios will be used when comparing candidate policies. Optimization scenarios will be separated from held-out validation scenarios so that a policy is not selected only because it performed well on one set of random draws.

3.5 Integrated Multi-objective Problem in Negative-Null Form

To place the three performance measures on comparable scales, fixed reference values W_{ref}, U_{ref}, and C_{ref} are used to normalize truck waiting time, mill shortfall, and yard cost, respectively. These

reference values will be carefully chosen and held fixed when comparing policies. Each deterministic or stochastic evaluation produces three normalized system measures:

Symbol	Definition	Units
W_{ref}	Reference truck-wait value	min
U_{ref}	Reference production-shortfall or backlog value	m^3
C_{ref}	Reference yard-cost value	\$/shift
λ	Fixed Pareto-preference vector	–
d	Physical-design vector	–
ζ	Deterministic truck-schedule and demand scenario	–
ξ	Random realization of truck supply	–
R	Number of stochastic simulation replications	integer
$N_L^{\min}, N_L^{\max}$	Lower and upper bounds on loader count	loaders

$$f(d,\theta) = \begin{bmatrix} W_{\text{truck}}/W_{\text{ref}} \\ U_{\text{mill}}/U_{\text{ref}} \\ C_{\text{yard}}/C_{\text{ref}} \end{bmatrix}. \tag{41}$$

For a selected preference vector λ,

$$J_\lambda(d,\theta) = \lambda^T f(d,\theta), \qquad \lambda_i \geq 0, \qquad \sum_i \lambda_i = 1. \tag{42}$$

The vector λ is a Pareto-analysis parameter and is not optimized. Different fixed values of λ describe different preferences among truck waiting, mill shortfall, and yard cost.

For the deterministic stages, the integrated problem is

$$
\begin{aligned}
\underset{d,\theta}{\text{minimize}} \quad & J_\lambda(d,\theta;\zeta) \\
\text{subject to} \quad & N_L^{\min} - N_L \le 0, \\
& N_L - N_L^{\max} \le 0, \\
& N_L \in \mathbb{Z}, \\
& -\theta_r \le 0 && \forall r, \\
& \sum_r \theta_r - 1 = 0, \\
& -I_s^P(t_k) \le 0, \\
& I_s^P(t_k) - C_s^P \le 0, \\
& -I_s^F(t_k) \le 0, \\
& I_s^F(t_k) - C_s^F \le 0, \\
& \sum_j z_{\ell j,k} - A_{\ell,k} \le 0, \\
& \sum_\ell z_{\ell j,k} - 1 \le 0, \\
& q - Q_\ell \le 0, \\
& W_{0.95} - \overline{W}_{0.95} \le 0, \\
& U_{\text{mill}} - \overline{U}_{\text{mill}} \le 0, \\
& \sum_m y_{sm} - 1 = 0, \\
& \sum_s y_{sm} - 1 \le 0, \\
& x_{k+1} - F(x_k, \pi(x_k;\theta); d, \zeta) = 0.
\end{aligned}
\tag{43}
$$

In Stage 1, d is fixed and only θ is optimized. In Stage 2, N_L and y_{sm} are released. In Stage 3, the same physical and policy constraints apply to every simulation replication, while the objective becomes the mean-variance problem in Eq. (38).

The event sequence makes J_λ discontinuous with respect to many variables. A small weight change can select a different task and change later queues and inventories. The study will therefore begin with grid, random, or other derivative-free searches rather than assume a useful gradient [1].

4 Dispatch Policy

This section defines the feasible task set, the four policy variables, the feature groups used to score tasks, and the conditions under which a loader remains idle.

4.1 Tasks and Assignment Rule

A baseline task is

$$j = (o(j), d(j), s(j), q)\,, \tag{44}$$

where $o(j)$ is the source, $d(j)$ is the destination, $s(j)$ is the species, and q is the fixed batch size. The task set is

$$\mathcal{J}(x_k) = \mathcal{J}^{T\to P}(x_k) \cup \mathcal{J}^{P\to F}(x_k) \cup \mathcal{J}^{T\to F}(x_k) \cup \{\text{idle}\}. \tag{45}$$

The three task subsets have the following meanings:

- $\mathcal{J}^{T\to P}(x_k)$ contains transfers from an arrived truck to the piles. The truck must contain at least q units of the task's species and the destination pile must have at least q units of free capacity.
- $\mathcal{J}^{P\to F}(x_k)$ contains transfers from a pile to the mill infeed. The pile must contain at least q units of the task's species and the infeed must have at least q units of free capacity.
- $\mathcal{J}^{T\to F}(x_k)$ contains direct transfers from an arrived truck to the mill infeed. The truck and infeed must satisfy the corresponding quantity and capacity constraints, and direct transfer must be permitted by $\chi_{TF} = 1$.

Each subset contains candidate material-transfer tasks, not loader assignments. A loader-task pair becomes feasible only when the task is in one of these subsets and the loader can reach and execute it.

The direct set $\mathcal{J}^{T\to F}$ is empty when $\chi_{TF} = 0$. All other infeasible tasks are removed before scoring.

For each available loader-task pair,

$$P_{\ell j}(x_k;\theta) = \theta_T \phi^T_{\ell j} + \theta_M \phi^M_{\ell j} + \theta_I \phi^I_{\ell j} - \theta_C \phi^C_{\ell j}. \tag{46}$$

The dispatcher assigns tasks sequentially within a decision event. It selects the compatible loader-task pair with the largest score, removes that loader and task from further consideration, updates affected features, and repeats until no unassigned loader has a positive-score task. Physical inventories change when transfers are completed, while material and capacity reservations are updated immediately after assignment so that later assignments at the same event do not reuse the same material or infeed capacity.

In the event of a tie, the dispatcher selects the loader-task pair with the shortest travel distance $\delta_{\ell j,k}$. If there is still a tie, the dispatcher selects the pair with the smallest loader index.

4.2 Four Policy Variables

The four weights have the following meanings:

- θ_T: truck waiting and inbound queue urgency;
- θ_M: mill-continuity and time-to-stockout urgency;
- θ_I: pile-inventory balance;
- θ_C: travel, active time, and immediate operating cost.

The weights are normalized by Eq. (30). The feature values change with the state, so one fixed policy can behave truck-first in one state and mill-first in another.

4.2.1 Feature Parameters and Derived Quantities

Symbol	Definition	Units
$\mathcal{J}_\ell(x_k)$	Tasks physically feasible for loader ℓ at event k	–
W_{ref}	Reference truck-wait time for normalization	min
$N_{T,\text{ref}}$	Reference truck-queue length for normalization	trucks
T_{ref}	Reference time scale for mill urgency	min
$A_s^{\text{comm}}(\tau)$	Committed future delivery volume of species s to the infeed over horizon τ	m^3
L_s^P	Nominal pile-replenishment lead time for species s	min
S_s^P	Safety-stock volume for species s in the pile	m^3
$I_s^{P,\text{tar}}(t)$	Desired pile inventory of species s	m^3
$C_s^{P,\text{free}}(t)$	Unreserved free capacity in the pile for species s	m^3
$\Delta I_s^P(j)$	Change in pile inventory caused by task j	m^3
ω_s^P	Species weight in the pile-inventory feature	–
Z_P	Normalization constant for the pile-inventory feature	m^3
α_1, α_2	Truck-feature coefficients, with $\alpha_1 + \alpha_2 = 1$	–
β_1, β_2	Mill-feature coefficients	–
$C_{\text{task,ref}}$	Reference task cost for normalization	\$/task
$\widehat{T}_s^{\text{res}}$	Estimated residence time of a newly stored batch of species s	min

4.3 Initial Feature Definitions

Each feature is normalized to a comparable range. A feature is zero when it does not apply to the task. For example, a pile-to-infeed task has no truck-wait feature, while a direct truck-to-infeed task can have both truck and mill urgency.

4.3.1 Truck Urgency

For truck b,

$$\widetilde{w}_b = \min\left\{1, \frac{t_k - a_b}{W_{\text{ref}}}\right\}. \tag{47}$$

Let $n_T(t_k)$ be the truck queue length. An initial feature is

$$\phi^T_{\ell j} = \mathbb{I}_{\{o(j)=b\}}\left[\alpha_1 \widetilde{w}_b + \alpha_2 \min\left\{1, \frac{n_T(t_k)}{N_{T,\text{ref}}}\right\}\right], \tag{48}$$

where $\alpha_1+\alpha_2 = 1$. The coefficients are fixed feature-definition parameters, not additional optimized policy weights in the first study.

4.3.2 Mill Urgency

Using current infeed inventory and already committed deliveries, define estimated time to stockout as

$$T_s^{\text{out}} = \inf\left\{\tau \geq 0 : I_s^F(t_k) + A_s^{\text{comm}}(\tau) - \int_{t_k}^{t_k+\tau} D_s(v)\, dv \leq 0\right\}. \tag{49}$$

This is the earliest future duration τ from the current decision time t_k at which the available/committed infeed inventory of species s is zero or less after accounting for mill consumption.

The species urgency is

$$u_s^M = \left[1 - \frac{T_s^{\text{out}}}{T_{\text{ref}}}\right]_+. \tag{50}$$

For a task delivering species s to the infeed,

$$\phi^M_{\ell j} = \mathbb{I}_{\{d(j)=F_s\}}\left[\beta_1 u_s^M + \beta_2\left[\frac{\tau_{\ell j,k} - T_s^{\text{out}}}{T_{\text{ref}}}\right]_+\right]. \tag{51}$$

The second term increases urgency when the task may finish after the predicted stockout.

4.3.3 Inventory Balance

The baseline inventory-balance feature is applied only to pile inventories. Let $I_s^{P,\mathrm{tar}}$ be the target pile inventory for species s.

For the initial model, the pile target is

$$I_s^{P,\mathrm{tar}}(t) = \max\left\{0,\, \min\left\{C_s^P - R_s^{P,\mathrm{comm}}(t),\, \int_t^{t+L_s^P} D_s(\tau)\, d\tau + S_s^P\right\}\right\}. \tag{52}$$

The demand integral represents the amount of species s expected to be consumed during a nominal pile-replenishment lead time L_s^P. The safety-stock term S_s^P provides additional protection against variation in truck arrivals, unloading time, or demand. The term $R_s^{P,\mathrm{comm}}(t)$ is the volume of species s already committed to enter the pile, such as material reserved from trucks assigned to pile unloading. Therefore, $C_s^P - R_s^{P,\mathrm{comm}}(t)$ is an upper bound on the current pile inventory that leaves room for the committed incoming material; it is not the current available space. The actual remaining unreserved space is

$$C_s^{P,\mathrm{free}}(t) = C_s^P - I_s^P(t) - R_s^{P,\mathrm{comm}}(t). \tag{53}$$

The minimum in Eq. (52) selects the smaller of the desired demand coverage and this space-based upper bound, while the outer maximum prevents the target from becoming negative. Feasibility constraints use the actual inventory $I_s^P(t)$ and therefore must subtract it when checking whether another batch can be accepted.

Initially, L_s^P and S_s^P are fixed scenario parameters, while $R_s^{P,\mathrm{comm}}(t)$ is updated whenever a pile-transfer task is committed. The target is therefore a desired operating level, not the physical pile capacity C_s^P. It increases when more reserve inventory is needed and decreases when incoming material already occupies the available pile space.

The pile-inventory change caused by task j is

$$\Delta I_s^P(j) = q\mathbb{I}_{\{d(j)=P_s\}} - q\mathbb{I}_{\{o(j)=P_s\}}. \tag{54}$$

The inventory feature measures the reduction in normalized pile-target deviation:

$$\phi_{\ell j}^I = \mathrm{clip}_{[0,1]}\left\{\frac{1}{Z_P}\sum_s \omega_s^P\left(|I_s^P - I_s^{P,\mathrm{tar}}| - |I_s^P + \Delta I_s^P(j) - I_s^{P,\mathrm{tar}}|\right)\right\}. \tag{55}$$

This feature rewards unloading a needed species into an under-target pile and can reward a pile-to-infeed task when a pile is above its target. In the baseline, infeed inventory is handled by the mill-urgency feature through time to stockout. An optional longer-horizon infeed feature ϕ_F^I could

be added later to compare infeed inventory with a desired multi-period buffer, but it is omitted initially because it would overlap with ϕ^M.

4.3.4 Operating Cost

The task cost depends on the loader, route, and destination:

$$c_{\ell j}^{\text{task}} = c_D \delta_{\ell j,k} + c_A \tau_{\ell j,k} + \mathbb{I}_{\{j \in \mathcal{J}^{T \to P}\}} h_{s(j)} q \, \widehat{T}_{s(j)}^{\text{res}}, \tag{56}$$

where $\widehat{T}_s^{\text{res}}$ is an initial estimate of pile residence time. The normalized feature is

$$\phi_{\ell j}^{C} = \min \left\{ 1, \frac{c_{\ell j}^{\text{task}}}{C_{\text{task,ref}}} \right\}. \tag{57}$$

The complete-shift holding cost is still calculated exactly through Eq. (17); the residence-time estimate only helps the local policy account for the likely consequence of storing a new batch.

4.4 Example Dispatch Decision

Consider one idle loader and the following state:

- a spruce truck has waited 20 minutes;
- the spruce infeed will be empty in approximately 6 minutes;
- the spruce pile is below its target inventory;
- direct truck-to-infeed service is allowed;
- three tasks are feasible: truck to pile, truck directly to infeed, and pile to infeed.

Suppose the normalized weights are

$$\theta = [0.25,\ 0.40,\ 0.15,\ 0.20]^T. \tag{58}$$

Illustrative feature values and scores are shown in Table 7. These values are examples, not calibrated data.

The direct task is selected because it serves the waiting truck and protects the mill. Its inventory feature is zero because it bypasses the pile. The pile-to-infeed task also receives zero inventory benefit because the spruce pile is below its target. If direct service were disabled, the pile-to-infeed task would be selected because it has the next-highest score.

Table 7: Illustrative feature calculation for one decision event.

Task	ϕ^T	ϕ^M	ϕ^I	ϕ^C	$P_{\ell j}$
Truck → spruce pile	0.80	0.00	0.70	0.20	0.265
Truck → spruce infeed	0.80	0.90	0.00	0.30	0.500
Spruce pile → infeed	0.00	0.90	0.00	0.15	0.330
Idle	0.00	0.00	0.00	0.00	0.000

4.5 Idle Decision

The baseline idle score is

$$P_{\ell,\text{idle}} = 0. \tag{59}$$

The loader performs a non-idle task only when

$$\max_{j \in \mathcal{J}_\ell(x_k) \setminus \{\text{idle}\}} P_{\ell j} > 0. \tag{60}$$

Idling is selected in three cases:

1. no non-idle task is physically feasible;
2. every feasible task has a negative score because cost exceeds its current urgency and inventory benefit;
3. the greatest non-idle score equals zero, in which case the tie is resolved in favor of idle to avoid an unnecessary movement.

An idle loader does not create an immediate zero-time decision loop. It waits until the next state-changing event, such as a truck arrival, another loader completing a task, an infeed threshold crossing, or a scheduled demand change. A short periodic review event may be added if continuous mill consumption can change urgency before another event occurs.

5 Conditional Extensions

This section defines optional additions to the baseline model. They will be implemented only if the initial results show that the added behavior could change the engineering conclusions.

5.1 Extension Nomenclature

Symbol	Definition	Units
$B_s(t)$	Production backlog of species s	m^3
$B_{s,0}$	Initial production backlog of species s	m^3
$B_{s,\mathrm{ref}}$	Reference backlog for policy-feature normalization	m^3
U_{backlog}	Terminal total production backlog	m^3
$U_{\mathrm{backlog,time}}$	Time-integrated production backlog	$\mathrm{m}^3{\cdot}\mathrm{min}$
$\overline{U}_{\mathrm{backlog}}$	Maximum allowable terminal backlog	m^3
θ_B	Optional backlog-urgency policy weight	–
$c_{\ell s,k}$	Species-s cargo carried by loader ℓ after stop k	m^3
$p_{\ell s,k}$, $d_{\ell s,k}$	Species-s pickup and delivery at stop k	m^3
$\gamma_{\ell j}(t)$	Progress of loader ℓ on task j	–
$y_{\ell j,k}$	1 if task j is interrupted for loader ℓ at event k	0 or 1
Γ_{safe}	Set of safe task-progress boundaries for interruption	–
H_L	Rollout horizon	min
R_L	Number of rollout replications per candidate action	integer
$\ell(x,u)$	Rollout running loss	–
$\widehat{V}_T(x)$	Estimated terminal value at rollout horizon	–
e_c	Entry time of inventory cohort c	min
$a_c(t)$	Age of cohort c at time t	min
$q_c(t)$	Remaining volume of cohort c	m^3
$\mathcal{C}_s(t)$	Species-s cohorts present at time t	–
$\mathcal{A}_s(t)$	Accessible species-s cohorts at time t	–
c_s^{age}	Ageing-penalty coefficient for species s	–
A_s^{free}	Age threshold before quality penalty begins	min
γ_s	Ageing-penalty exponent for species s	–

5.2 Production Backlog and Backlog-Aware Dispatch

The baseline reports production shortfall as the amount of scheduled production that is lost when the mill is starved. A likely extension is to represent this quantity explicitly as a species-specific production backlog. Let $B_s(t)$ denote the amount of species-s production that has been requested but not completed.

A simple backlog balance is

$$\frac{dB_s(t)}{dt} = D_s(t) - r_s^{\text{proc}}(t), \qquad B_s(t) \geq 0, \qquad B_s(0) = B_{s,0}. \tag{61}$$

Backlog accumulates whenever requested production is not completed. Unlike the baseline shortfall calculation, the backlog may later be served using the same fixed processing rate. In particular, if the currently demanded species is unavailable but a backlogged species is present in the infeed, the mill may process the backlogged species instead. The backlog then decreases for that species without any increase in the mill rate. For a single mill, the extension also requires that at most one species be processed at a time:

$$\sum_{s \in \mathcal{S}} \mathbb{I}_{\{r_s^{\text{proc}}(t) > 0\}} \leq 1. \tag{62}$$

The explicit backlog state allows the dispatch policy and the mill sequencing rule to respond to accumulated production delay rather than only to current infeed inventory.

Backlog recovery does not increase the mill processing rate. The mill continues to process at the same fixed rate $\bar{r}_s$, but the species receiving that processing can be selected from either the current demand or an existing backlog. For example, if the currently demanded species is unavailable in the infeed while a species with $B_s(t) > 0$ is available, the mill can switch to the backlogged species. A similar switch can be made when serving the backlog is preferable to continuing with the current demand. The processing rate remains either zero or the applicable fixed rate throughout.

When this extension is activated, this backlog-aware sequencing rule replaces the baseline requirement that a species can be processed only when $D_s(t) = \bar{r}_s$.

The backlog objective replaces the unmet-demand objective rather than adding a separate backlog cost. A simple choice is the residual backlog at the end of the shift:

$$U_{\text{backlog}} = \sum_{s \in \mathcal{S}} B_s(H). \tag{63}$$

The optimization then minimizes $U_{\text{backlog}}/U_{\text{ref}}$ in place of $U_{\text{mill}}/U_{\text{ref}}$. If delayed production should also be penalized according to how long it remains outstanding, the objective can instead use the area under the backlog curve,

$$U_{\text{backlog,time}} = \sum_{s \in \mathcal{S}} \int_0^H B_s(t)\, dt. \tag{64}$$

The first implementation will use terminal backlog unless the results show that backlog duration changes the dispatch decisions.

When this extension is activated, U_{backlog} replaces U_{mill} in the integrated objective and $U_{\text{backlog}} - \overline{U}_{\text{backlog}} \leq 0$ replaces the baseline unmet-demand constraint.

The dispatch policy can use a simple backlog urgency feature in addition to time-to-stockout. For a candidate task j delivering species s to the infeed, one possible feature is

$$\phi^B_{\ell j} = \mathbb{I}_{\{d(j)=F_s\}} \operatorname{clip}_{[0,1]} \left(\frac{B_s(t_k)}{B_{s,\mathrm{ref}}} \right), \tag{65}$$

The first term increases priority when the candidate task supplies a species with a large outstanding backlog. The priority score could then become

$$P^{\mathrm{backlog}}_{\ell j} = P_{\ell j} + \theta_B \phi^B_{\ell j}, \qquad \theta_B \geq 0. \tag{66}$$

If backlog urgency is added as a separate policy component, the four-weight simplex is replaced by a five-weight simplex:

$$\theta_T + \theta_M + \theta_I + \theta_C + \theta_B = 1, \qquad \theta_r \geq 0. \tag{67}$$

For the initial implementation, $\phi^B_{\ell j}$ uses only the normalized backlog $B_s/B_{s,\mathrm{ref}}$. The mill-side rule determines whether the current demand or a backlogged species is served, while the loader policy determines which material is delivered to the infeed. This extension is expected to be added before multistage routes because it gives the policy a direct memory of earlier starvation events while preserving the existing task structure.

5.3 Multistage Routes and Task Preemption

The baseline task has one source and one destination. A multistage route would let a loader visit more than one source before reaching the infeed. Let $c_{\ell s,k}$ be the species-s cargo on loader ℓ. A stop-level action is

$$u_{\ell,k} = (n_{\ell,k+1}, \{p_{\ell s,k}\}, \{d_{\ell s,k}\}), \tag{68}$$

with cargo balance

$$c^+_{\ell s,k} = c^-_{\ell s,k} + p_{\ell s,k} - d_{\ell s,k}, \qquad \sum_s c^+_{\ell s,k} \leq Q_\ell. \tag{69}$$

Node inventories must decrease when material is picked up and increase when it is delivered. Route-continuity constraints must also ensure that each stop begins where the previous stop ended. To limit the action space, the first extension would allow only direct routes and routes with one intermediate pickup.

Task preemption would divide a transfer into safe stages: travel to source, loading, travel to destina-

tion, and unloading. Let $\gamma_{\ell j}(t)$ be task progress and let $y_{\ell j,k} = 1$ indicate interruption. Preemption is allowed only at a safe stage boundary:

$$y_{\ell j,k} - \mathbb{I}_{\{\gamma_{\ell j}(t_k) \in \Gamma_{\text{safe}}\}} \leq 0. \tag{70}$$

Interrupted work returns to the request queue, and a switching cost is added for lost travel or setup. This extension models emergency reassignment without allowing unsafe interruption during loading or unloading.

5.4 Expected Future Cost Rollout

The baseline policy uses current-state features and a time-to-stockout estimate. A rollout would simulate the short-term consequence of each feasible current action. For candidate task j,

$$\widehat{Q}_{H_L}(x_k, j) = \frac{1}{R_L} \sum_{r=1}^{R_L} \left[\int_{t_k}^{t_k + H_L} \ell(x^{(r,j)}(t), u^{(r,j)}(t)) dt + \widehat{V}_T(x^{(r,j)}(t_k + H_L)) \right]. \tag{71}$$

Each rollout forces task j first, then uses the baseline policy for later decisions. Future trucks in the rollout are sampled from the same arrival and composition distributions used by the main stochastic model. These samples are forecasts used to compare current actions. They do not need to match the truck arrivals that occur in the real simulation after the action is selected. The same sampled rollout futures should be used for every candidate action at one decision event so that the action comparison is less noisy.

The selected action is

$$j^* = \arg\min_j \widehat{Q}_{H_L}(x_k, j). \tag{72}$$

This extension requires temporary simulation states, a rollout horizon, a terminal-value estimate, and additional computation at every decision event.

5.5 Breakdowns and Robustness Sensitivity

An equipment-reliability extension would add loader states for available, busy, failed, and under service. Failure time and repair duration would be sampled from assumed or measured distributions. A first implementation would apply failures at task boundaries so that the baseline material-transfer model remains intact. A failed loader would leave the feasible assignment set until repair is complete.

The model would be evaluated under three cases: no failures, nominal availability, and degraded availability. Results would compare changes in mean performance, variance, tail truck waits, and

mill interruptions. If failure risk materially changes dispatch behavior, a loader-task reliability feature could be included in the operating-cost feature. No additional risk constraint is proposed until the baseline sensitivity results show that one is needed.

5.6 Ageing Inventory and Quality Loss

The baseline includes inventory holding cost but does not reduce log quality with time. An ageing extension would retain pile inventory in truck-sized or batch-sized cohorts. If cohort c enters pile storage at time e_c, its age is

$$a_c(t) = t - e_c. \tag{73}$$

Pile inventory becomes

$$I_s^P(t) = \sum_{c \in \mathcal{C}_s(t)} q_c(t). \tag{74}$$

A soft degradation cost could be

$$C_{\text{age}} = \sum_{s,c} \int_0^H q_c(t) c_s^{\text{age}} \left[a_c(t) - A_s^{\text{free}} \right]_+^{\gamma_s} dt, \tag{75}$$

where A_s^{free} is a period with no modelled quality penalty. A later model could impose a maximum acceptable age, but only if suitable data are available.

To avoid treating each log as a separate action, pile withdrawals would use an oldest-first rule among accessible cohorts of each species:

$$c_s^*(t) = \arg\max_{c \in \mathcal{A}_s(t)} a_c(t), \tag{76}$$

where $\mathcal{A}_s(t)$ is the set of accessible cohorts. The policy would still choose the species and transfer route. The pile rule would choose the cohort. This extension is most useful for a multi-day simulation or a one-day simulation initialized with inventory of different ages. It would also represent the tradeoff created by direct truck-to-infeed service: direct service may save handling now, while pile withdrawal may be preferable when older material is approaching a quality limit.

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